

University of Rwanda

College of science and technology

Computer engineering

Mobile application systems and design Lab#3

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TRAVEL APP — Layout Choices, Widgets Used & Navigation Flow

1. EXPLANATION OF LAYOUT CHOICES

A. Home Screen Layout

```
Column(  
  children: [  
    SizedBox(height: 50, child: SearchBar()),  
    SizedBox(height: 40, child: CategoriesList()),  
    SizedBox(height: 200, child: PopularList()),  
    Expanded(child: DestinationsGrid()),  
  ],  
)
```

Why this layout?

- **Fixed heights** : Prevents overflow and maintains UI stability
- **SizedBox** : Gives search bar, categories, and popular list consistent height
- **Expanded** : Allows destination grid to fill remaining space and scroll naturally
- **Column** : Simple, predictable vertical arrangement

B. Detail Screen Layout

```
CustomScrollView(
  slivers: [
    SliverAppBar(expandedHeight: 200),
    SliverToBoxAdapter(
      child: Padding(
        padding: EdgeInsets.all(16),
        child: Column(...),
      ),
    ),
  ],
)
```

Why this layout?

- **CustomScrollView + SliverAppBar** → Enables a professional collapsing header animation
- **SliverToBoxAdapter** → Combines slivers with normal widgets
- **Padding(16)** → Consistent spacing, improves readability
- **Column** → Organizes information smoothly (title, description, price, CTA button)

C. Booking Screen Layout

```
SingleChildScrollView(
  padding: EdgeInsets.all(16),
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [...],
  ),
)
```

Why this layout?

- **SingleChildScrollView** → Ensures forms remain scrollable on small devices
- **CrossAxisAlignment.start** → Proper left alignment for form labels
- **Consistent 16px padding** → Matches Material Design spacing
- **Cards** → Groups related sections visually (destination details, price summary)

2. WIDGETS USED (41 Total)

Category	Widgets Used	Count
Layout	Scaffold, Column, Row, Container, Padding, SizedBox, Expanded, Stack, Positioned	9
Scroll	ListView, GridView, SingleChildScrollView, CustomScrollView, SliverAppBar, SliverToBoxAdapter	6
Navigation	Navigator, MaterialPageRoute, GestureDetector, InkWell	4
Display	Text, Image, Icon, Card, AppBar, BottomNavigationBar	6
Input	TextField, TextFormField, ElevatedButton, IconButton, DropdownButton	5
Feedback	CircularProgressIndicator, SnackBar, AlertDialog	3
Styling	Theme, BoxDecoration, Border, BorderRadius, BoxShadow	5
Utility	SafeArea, AspectRatio, ClipRRect	3
TOTAL	41 widgets	

Widget Usage Examples

Stack with Positioned (Image Overlay Icon)

```
Stack(  
  children: [  
    Image.asset('image.jpg'),  
    Positioned(  
      bottom: 10,  
      right: 10,  
      child: Icon(Icons.favorite),  
    ),  
  ],  
)
```

Card with rounded corners

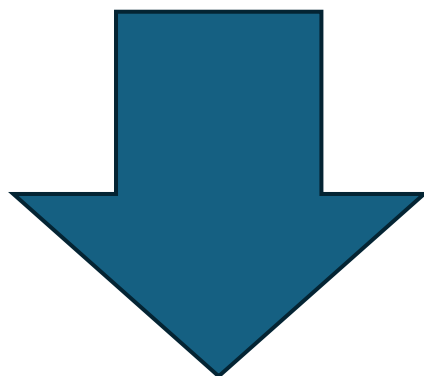
```
Card(  
  shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(15)),  
  child: Column(...),  
)
```

CustomScrollView with Slivers

```
CustomScrollView(  
  slivers: [  
    SliverAppBar(...),  
    SliverToBoxAdapter(...),  
  ],  
)
```

3. NAVIGATION FLOW

Below navigation flow diagram



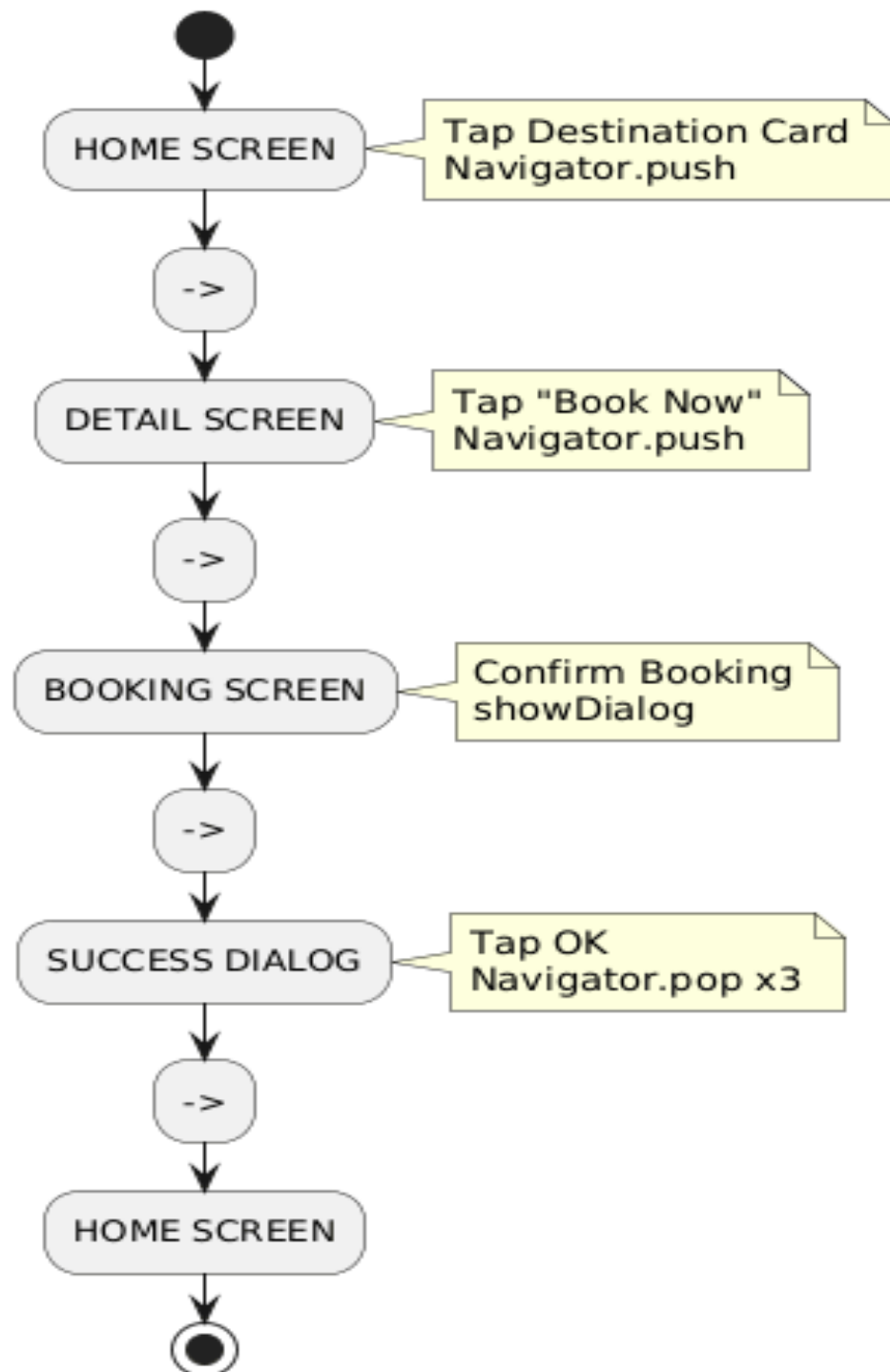


Fig:1.1

Navigation Code Examples

Forward Navigation

```
Navigator.push(  
  context,  
  MaterialPageRoute(  
    builder: (context) => DetailScreen(destination: destination),  
  ),  
);
```

Back Navigation

```
Navigator.pop(context);
```

Passing Data Between Screens

```
class DetailScreen {  
  final Destination destination;  
  const DetailScreen({required this.destination});  
}
```

Flow Description

1. *Home → Detail:*
User taps any destination card.
2. *Detail → Booking:*
User taps “Book Now”.
3. *Booking → Success Dialog:*
User taps Confirm, dialog appears.
4. *Dialog → Home:*
Press OK, pops back 3 screens.
5. *Back Button:*
Each screen pops to the previous one naturally.

(All screenshots of the Home Screen, Detail Screen, Booking Screen, and Success Dialog are already included in the same project repository for reference.)